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85/100

## TRANSPORT PHENOMENA I

FALL 2025

## EXAM 1

A thin liquid film of thickness  $\delta$  is flowing down on the inner surface of a long vertical cylindrical tube of inner radius  $R$ .

Assumptions:

- Laminar flow
- Newtonian fluid with constant density  $\rho$  and constant viscosity  $\mu$
- Constant pressure within the liquid:  $\frac{\partial p}{\partial z} = 0$  *→ there is no pressure*
- Steady state flow, where  $v_z = v_z(r)$  and  $v_r = v_\theta = 0$

Answer the following questions, in terms of parameters  $\delta$ ,  $R$ ,  $\mu$ ,  $\rho$ , and  $g$ .

- 1- Write down the steady state  $z$ -momentum balance on a cylindrical shell of thickness  $\Delta r$  and length  $L$  (represented as the dashed lines in the schematics).

- 2- Apply Newton's law of viscosity to the above momentum balance to derive a second-order differential equation for the velocity profile of the liquid.

- 3- Write down two boundary conditions for this velocity profile.

- 4- Derive the velocity profile by integrating the differential equation of part 2 and evaluating the integration constants.

- 5- Calculate the mass flow rate of the liquid film.

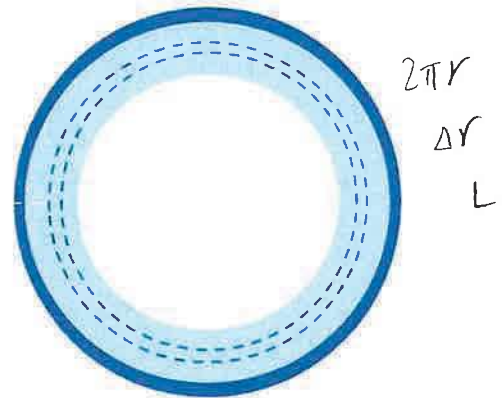
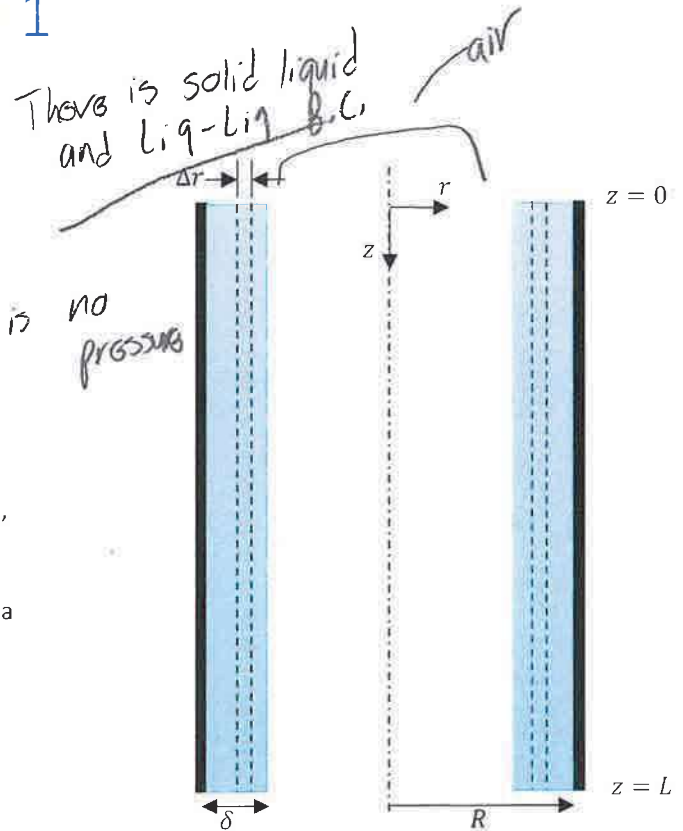
Hint:  $\int dx x \log x = \frac{1}{2} x^2 \log x - \frac{1}{4} x^2$

$\downarrow$   
 $\ln$

$$\int \delta \rho v$$

$$\int_0^R dr \int_0^{2\pi} r d\theta$$

$$r \ln r = \frac{1}{2} x^2 \ln r - \frac{1}{4} r^2$$





Problem 1Momentum balance

$$\left[ \text{Rate of convective momentum in} \right] - \left[ \text{rate of convective momentum out} \right] + \left[ \text{rate of molecular momentum in} \right] - \left[ \text{rate of molecular momentum out} \right] + \text{gravity} = 0$$

$$\text{convective} = \rho |V| < V | \rightarrow \rho V_z^2 \text{ b/c only } V_z \text{ exists}$$

$$\text{molecular} = |\tau| + P \rightarrow \tau_{zr}, \text{ and pressure will cancel b/c } \frac{\partial P}{\partial z} = 0$$

$$\text{gravity} = \rho(2\pi r \Delta r L)g$$

$$\left[ 2\pi r \Delta r \rho V_z^2 \right]_{z=0} - \left[ 2\pi r \Delta r \rho V_z^2 \right]_{z=L} \left. \begin{array}{l} \text{convective momentum balance} \\ \text{these terms cancel out b/c steady} \\ \text{laminar flow} \end{array} \right\}$$

$$\left[ 2\pi r L \tau_{zr} \right]_{r=r+\Delta r} - \left[ 2\pi r L \tau_{zr} \right]_{r=r} \left. \begin{array}{l} \text{viscous momentum balance} \end{array} \right\}$$

$$\left[ 2\pi r \Delta r P \right]_{z=0} - \left[ 2\pi r \Delta r P \right]_{z=L} \left. \begin{array}{l} \text{pressure forces which cancel} \\ \text{out} \end{array} \right\}$$

$$\left[ \underbrace{\rho(2\pi r \Delta r L)}_{\text{mass}} \underbrace{g}_{\text{force}} \right] \left. \begin{array}{l} \text{Force of gravity} \end{array} \right\}$$

Total momentum balance

$$\left[ 2\pi r L \tau_{zr} \right]_{r=r} - \left[ 2\pi r L \tau_{zr} \right]_{r=r+\Delta r} + \rho(2\pi r \Delta r L)g = 0$$

## Problem 2

$$[2\pi r L \tau_{zr}]_{r=r} - [2\pi r L \tau_{zr}]_{r=r+\Delta r} + \rho 2\pi r \Delta r L g = 0$$

$$\tau_{zr} = -\mu \frac{dv_z}{dr}$$

\* when dividing by volume little  $r$  does not cancel \*

$$[2\pi r L -\mu \frac{dv_z}{dr}]_{r=r} - [2\pi r L -\mu \frac{dv_z}{dr}]_{r=r+\Delta r} + \rho 2\pi r \Delta r L g = 0$$

$$\frac{2\pi r \Delta r L}{r \Delta r} \left[ (r) - \mu \frac{dv_z}{dr} \right]_{r=r} - \frac{2\pi r \Delta r L}{r \Delta r} \left[ (r) - \mu \frac{dv_z}{dr} \right]_{r=r+\Delta r} + \rho g = 0$$

↓ This can equals

$$\frac{(r) - \mu \frac{dv_z}{dr}}{r \Delta r} \Big|_{r=r+\Delta r} - \frac{(r) - \mu \frac{dv_z}{dr}}{r \Delta r} \Big|_{r=r} = \rho g$$

$$\lim_{\Delta r \rightarrow 0} \left[ r \frac{dv_z}{dr} \right]_{r=r+\Delta r} - \left[ r \frac{dv_z}{dr} \right]_{r=r} = -\frac{\rho g}{\mu} r$$

$$\boxed{\frac{d}{dr} \left( r \frac{dv_z}{dr} \right) = -\frac{\rho g}{\mu} r} \leftarrow \text{2nd order differential Eqn.}$$

## Problem 3

- There is a solid liquid interface B.C. and a liquid air (liquid-liquid) interface B.C., because there is air in the middle of the pipe

$$\text{Solid Lq. B.C.} \rightarrow v_z(r=R) = 0$$

$$\text{Liq-Liq interface} \rightarrow \left. \frac{dv_z}{dr} \right|_{r=s} = 0$$

Problem 4

B.C.  $\rightarrow V_z(R) = 0$

$$\frac{d}{dr} \left( r \frac{dv_z}{dr} \right) = -\frac{\rho g}{\mu} r$$

$$\left. \frac{dv_z}{dr} \right|_{r=s} = 0$$

↓ first integration

$$\int \frac{d}{dr} \left( r \frac{dv_z}{dr} \right) = \int -\frac{\rho g}{\mu} r$$

divide both sides by r

$$r \frac{dv_z}{dr} = -\frac{\rho g}{2\mu} r^2 + C_1 \quad \uparrow \quad \frac{dv_z}{dr} = -\frac{\rho g}{2\mu} r + \frac{C_1}{r}$$

Evaluating 1st B.C.

$$\left. \frac{dv_z}{dr} \right|_{r=s} = 0 = -\frac{\rho g}{2\mu} s + \frac{C_1}{s} \quad \rightarrow \quad C_1 = \frac{\rho g}{2\mu} s^2$$

↓ now we have

$$\frac{dv_z}{dr} = -\frac{\rho g}{2\mu} r + \frac{\rho g s^2}{2\mu} \frac{1}{r}$$

↓ Integrate again

$$V_z = -\frac{\rho g}{4\mu} r^2 + \frac{\rho g s^2}{2\mu} \ln|r| + C_2$$

Evaluating 2nd B.C.

$$V_z(R) = 0 = -\frac{\rho g R^2}{4\mu} + \frac{\rho g s^2}{2\mu} \ln|R| + C_2$$

$$C_2 = \frac{\rho g R^2}{4\mu} - \frac{\rho g s^2}{2\mu} \ln|R|$$

$$V_z = -\frac{\rho g r^2}{4\mu} + \frac{\rho g s^2}{2\mu} \ln|r| + \frac{\rho g R^2}{4\mu} - \frac{\rho g s^2}{2\mu} \ln|R|$$

# Problem 5

Mass flow rate

$$\int_S ds \rho W \rightarrow \int ds \rho \left( \underbrace{-\frac{\rho g r^2}{4\mu}}_{-\frac{\rho g}{4\mu}(R^2 - r^2)} + \underbrace{\frac{\rho g s^2}{2\mu} \ln|r|}_{\frac{\rho g s^2}{2\mu} \ln\left(\frac{r}{R}\right)} + \frac{\rho g R^2}{4\mu} - \frac{\rho g s^2}{2\mu} \ln|R| \right)$$

$$\int_0^R dr \int_0^{2\pi} r d\theta \rho \left( -\frac{\rho g}{2\mu} \left( -\frac{r^2}{2} + \frac{R^2}{2} + s^2 \ln\left|\frac{r}{R}\right| \right) \right)$$

$$\rho \frac{\rho g}{2\mu} 2\pi \int_0^R r \left( -\frac{r^2}{2} + \frac{R^2}{2} + s^2 \ln\left|\frac{r}{R}\right| \right) dr$$

$$\frac{\rho^2 g 2\pi}{2\mu} \int_0^R \left( -\frac{r^3}{2} + \frac{R^2}{2} r + (s^2 \ln\left|\frac{r}{R}\right|) r \right) dr$$

$$\frac{\rho^2 g \pi}{\mu} \left[ -\frac{r^4}{8} + \frac{R^2}{4} r^2 + s^2 \left( \frac{1}{2} \left( \frac{r}{R} \right)^2 \ln\left|\frac{r}{R}\right| - \frac{1}{4} \left( \frac{r}{R} \right)^2 \right) \right] \Big|_0^R$$

$$\frac{\rho^2 g \pi}{\mu} \left( -\frac{R^4}{8} + \frac{R^2}{4} R^2 + s^2 \left( \frac{1}{2} \frac{R^2}{R^2} \ln\left|\frac{R}{R}\right| - \frac{1}{4} \left( \frac{R^2}{R^2} \right) \right) \right)$$

← unsimplified mass flow rate.

$$\frac{\rho^2 g \pi}{\mu} \left( -\frac{R^4}{8} + \frac{R^4}{4} - \frac{s^2}{4} \right) = \boxed{\frac{\rho^2 g \pi}{\mu} \left( \frac{R^4}{8} - \frac{s^2}{4} \right)}$$

↑ mass flow rate simplified